Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

School of Computer Science Engineering & Information Systems (SCORE)

WINTER SEMESTER 2025-2026

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Registration Number: 23MIS0174

Course Title: Agile Development Process and Devops Lab

Course Code: ISWE406P

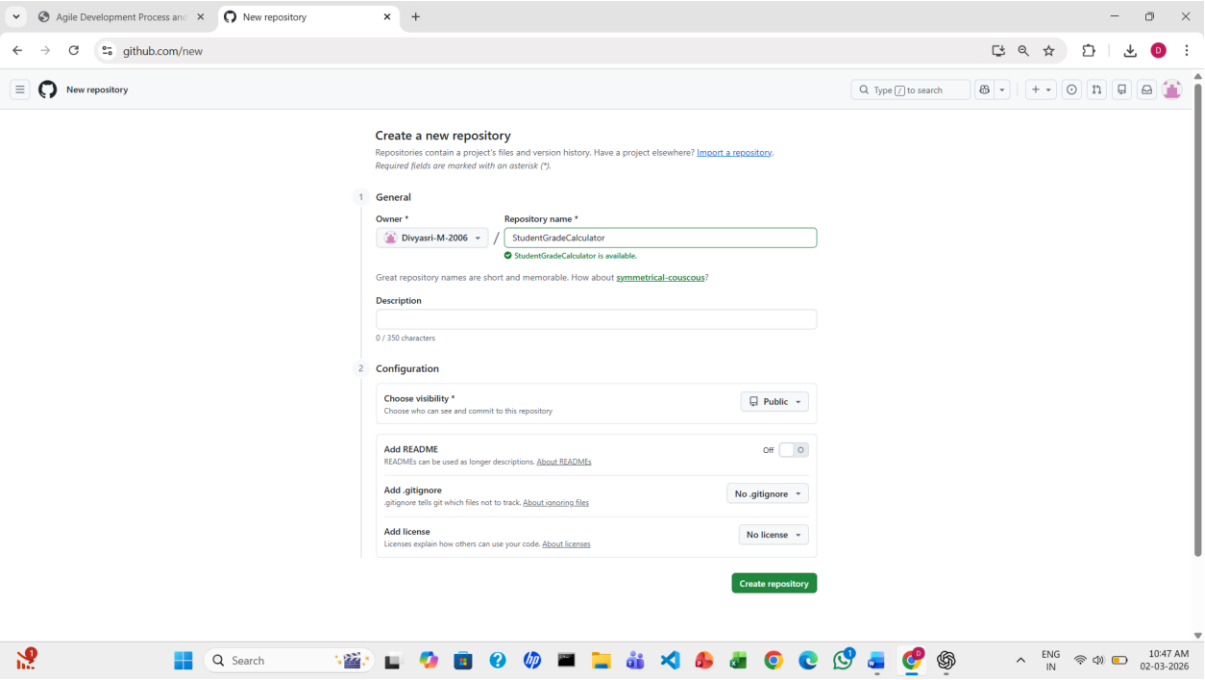
Faculty: Dr. Sweta Bhattacharya

Slot: L3+L4

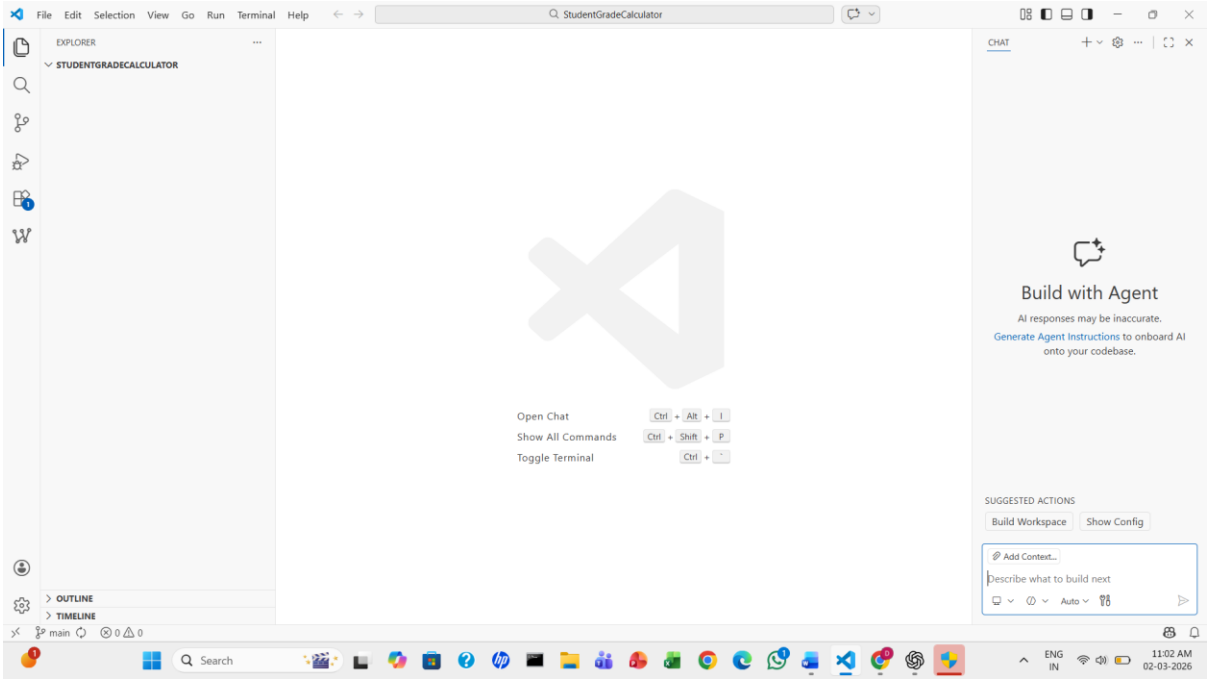
LAB ASSIGNMENT – 3

1. Create a Maven Project in Git

- Create a new Git repository



- Clone the Repository Locally

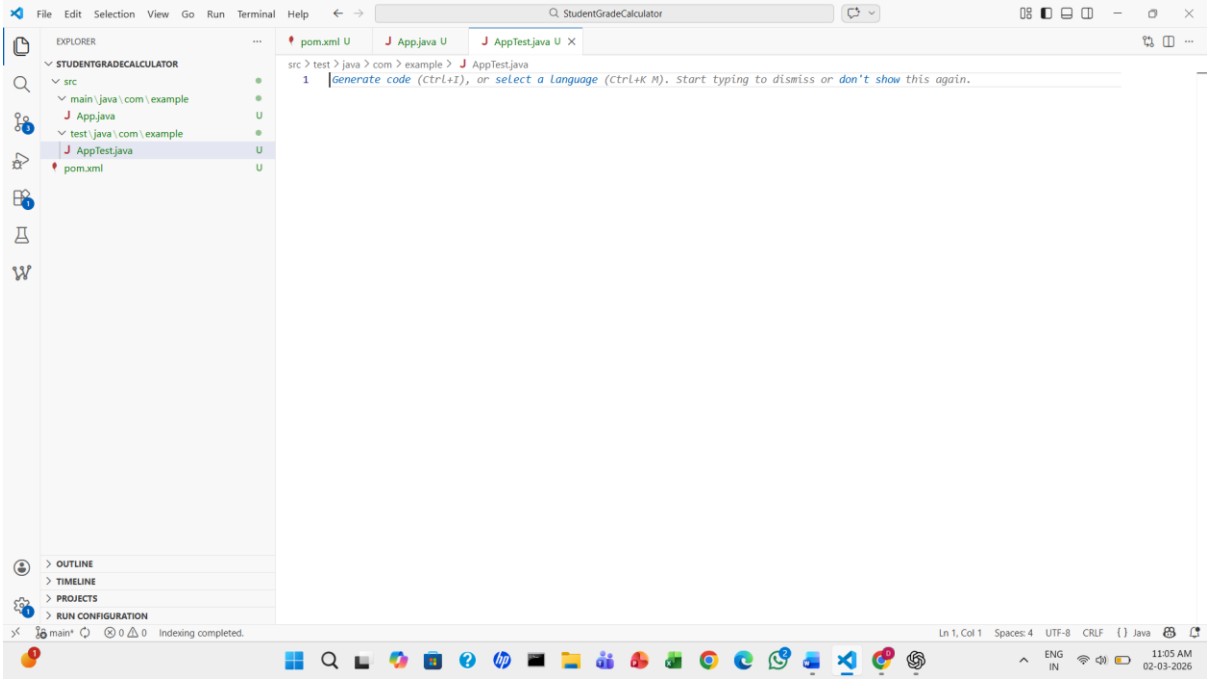


- Create a simple Maven project structure manually, including the following files and folders:

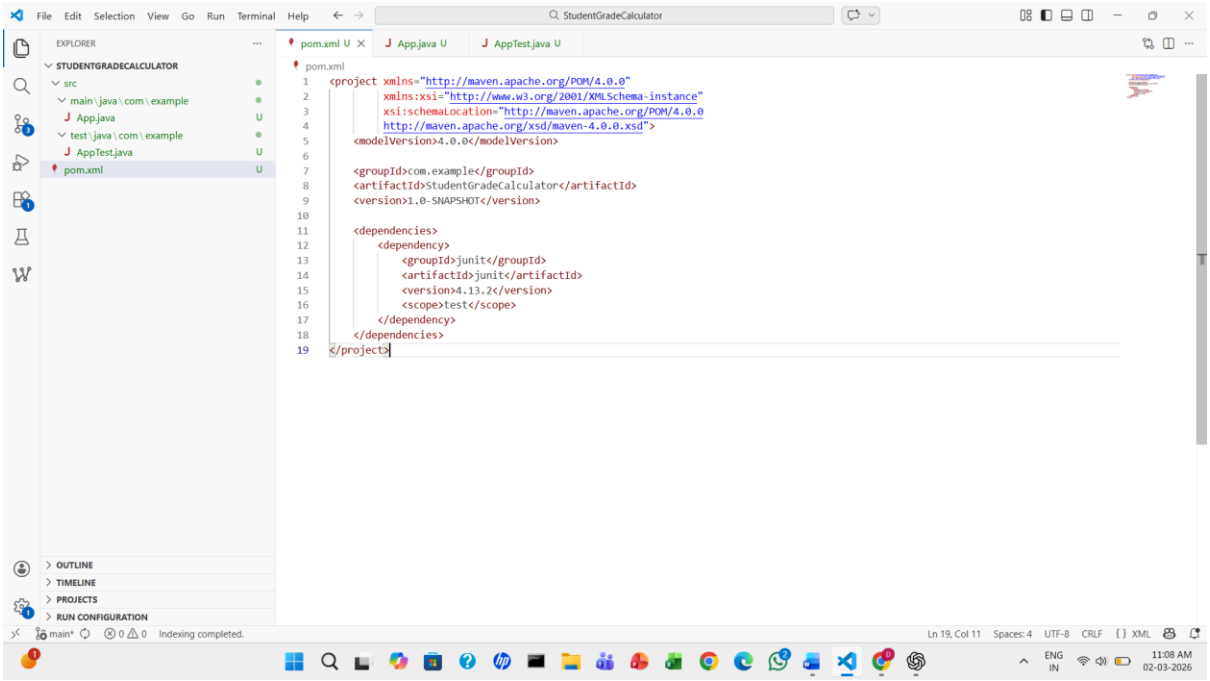
  pom.xml

  src/main/java/com/example/App.java

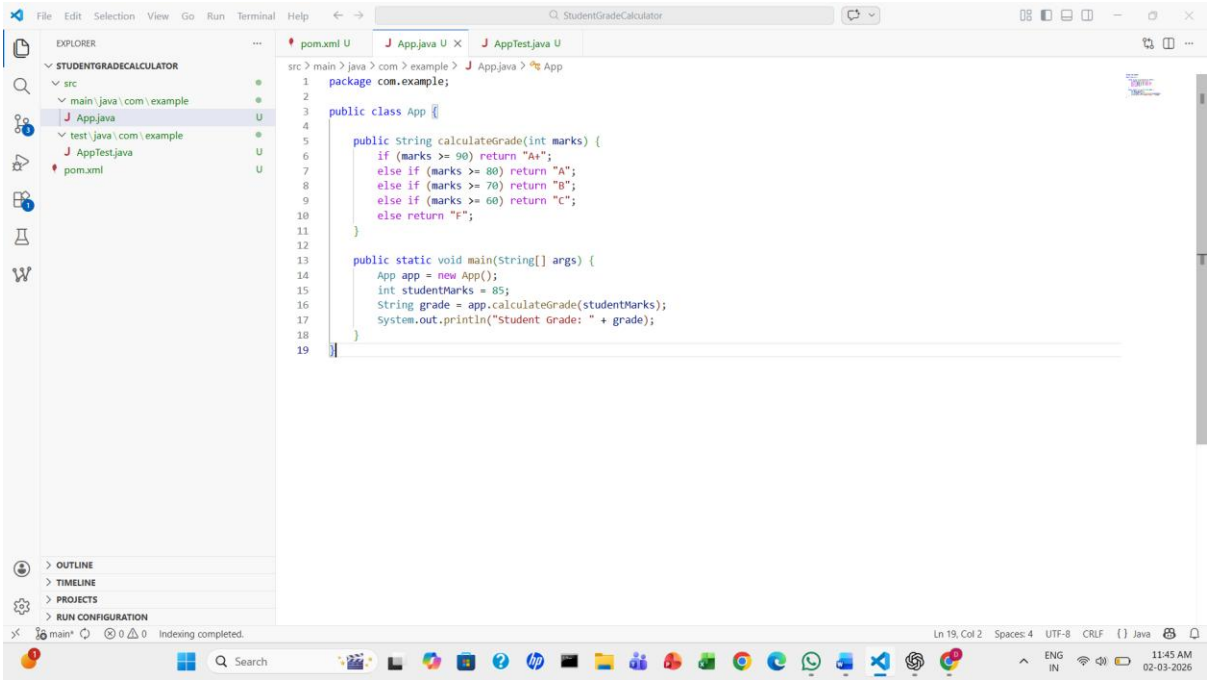
  src/test/java/com/example/AppTest.java



2. Implement the Java Program

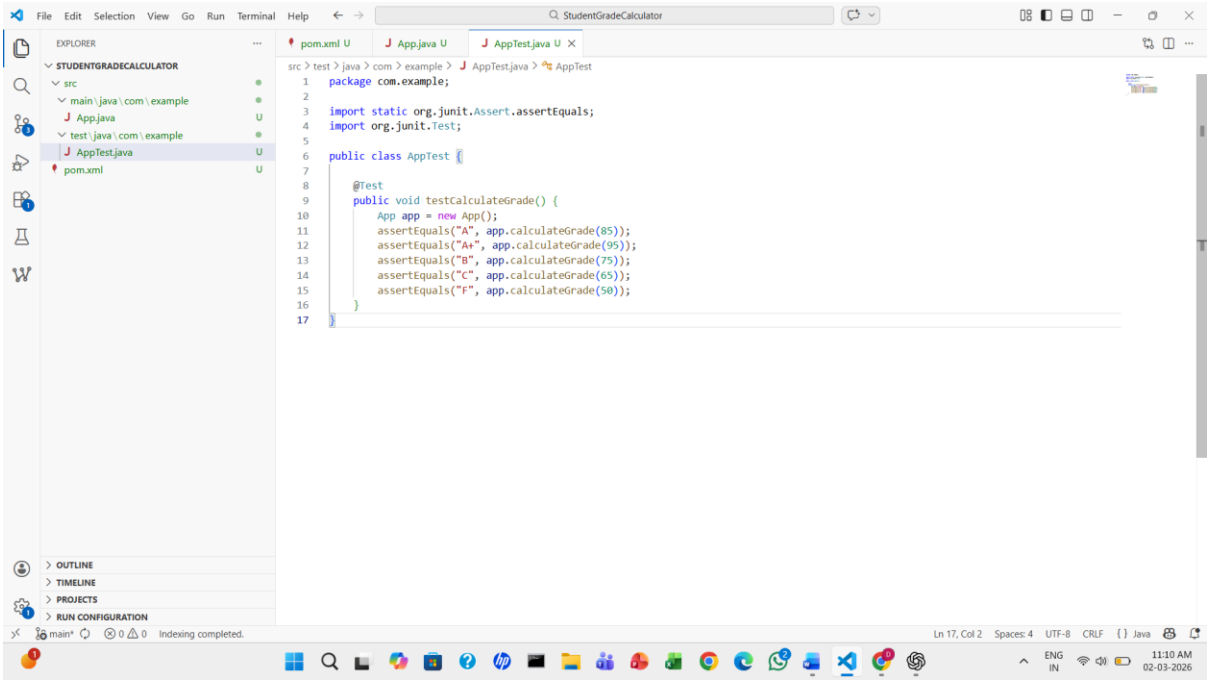


- Write a simple Java program in App.java under your newly created Git repository for Student Grade Calculator:



```java
package com.example;

public class App {

    public String calculateGrade(int marks) {
        if (marks >= 90) return "A+";
        else if (marks >= 80) return "A";
        else if (marks >= 70) return "B";
        else if (marks >= 60) return "C";
        else return "F";
    }

    public static void main(String[] args) {
        App app = new App();
        int studentMarks = 85;
        String grade = app.calculateGrade(studentMarks);
        System.out.println("Student Grade: " + grade);
    }
}
```

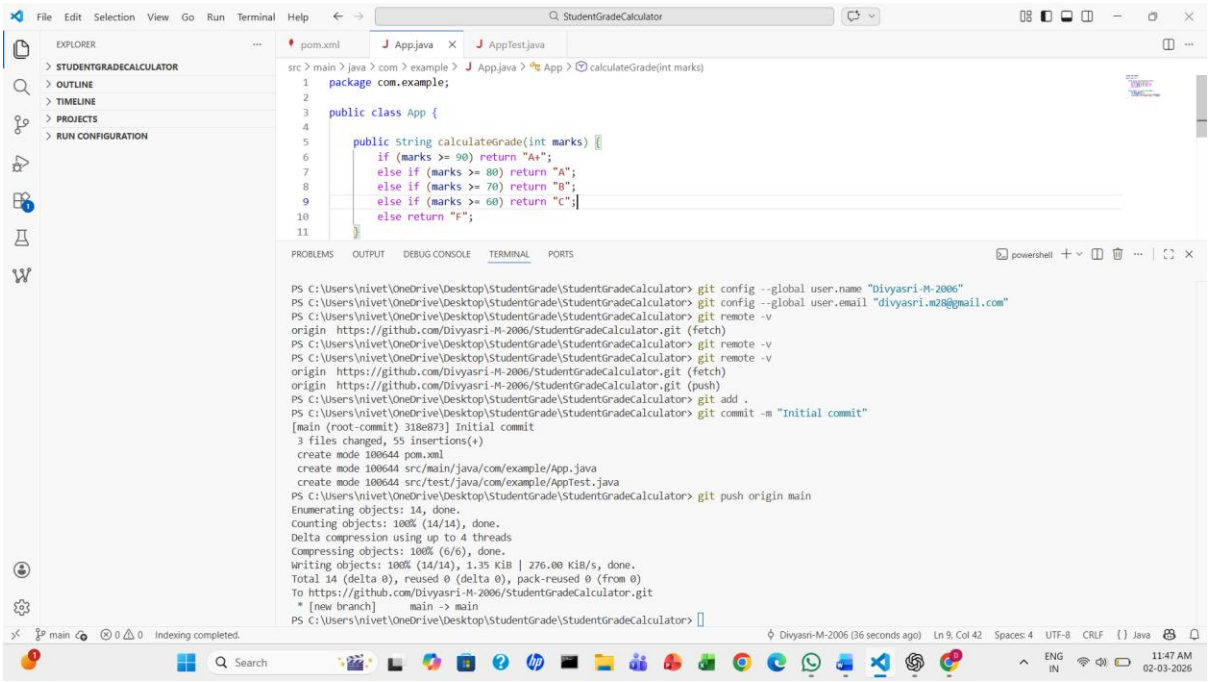
- Write a corresponding unit test in AppTest.java.



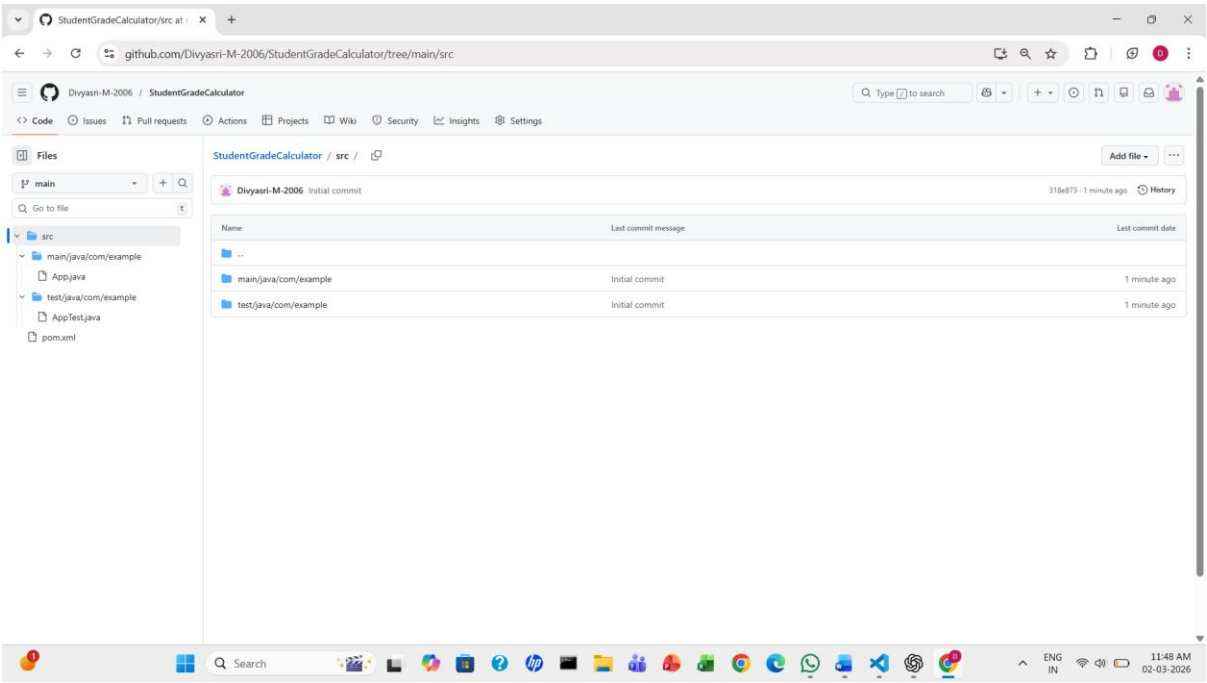
```java
package com.example;

import static org.junit.Assert.assertEquals;
import org.junit.Test;

public class AppTest {

    @Test
    public void testCalculateGrade() {
        App app = new App();
        assertEquals("A", app.calculateGrade(85));
        assertEquals("A+", app.calculateGrade(95));
        assertEquals("B", app.calculateGrade(75));
        assertEquals("C", app.calculateGrade(65));
        assertEquals("F", app.calculateGrade(50));
    }
}
```

3. Push and Clone the Repository

- Clone the repository using Visual Studio Code.



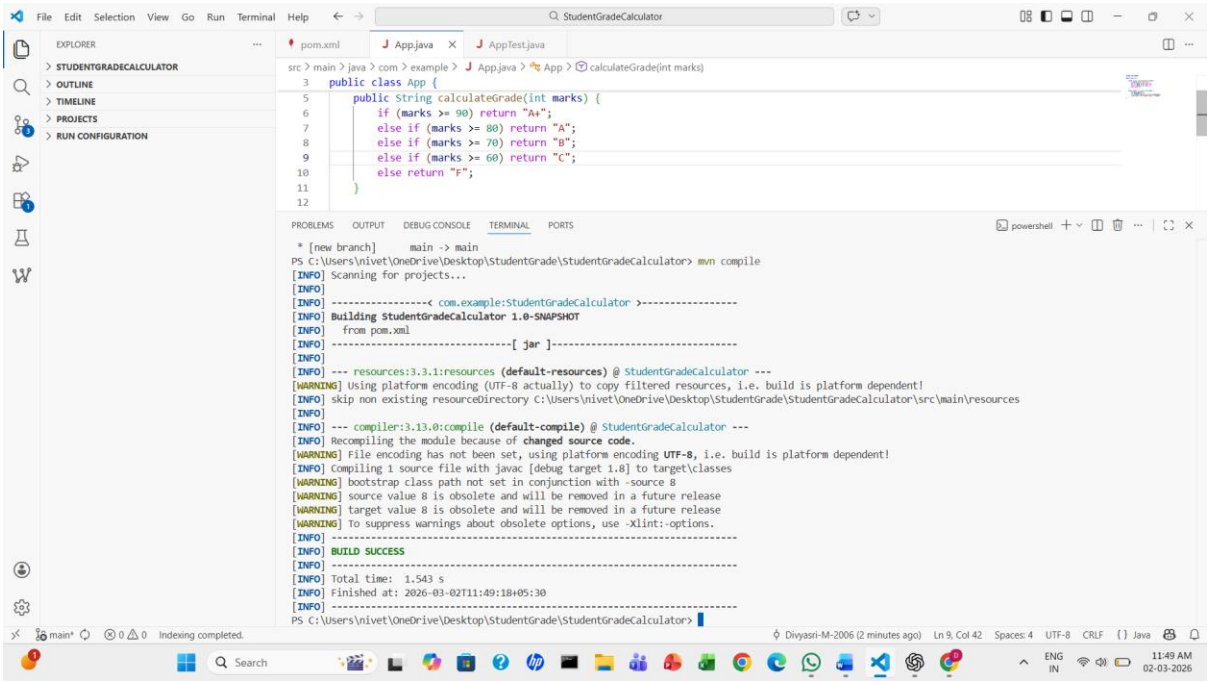
- Verify the project structure.



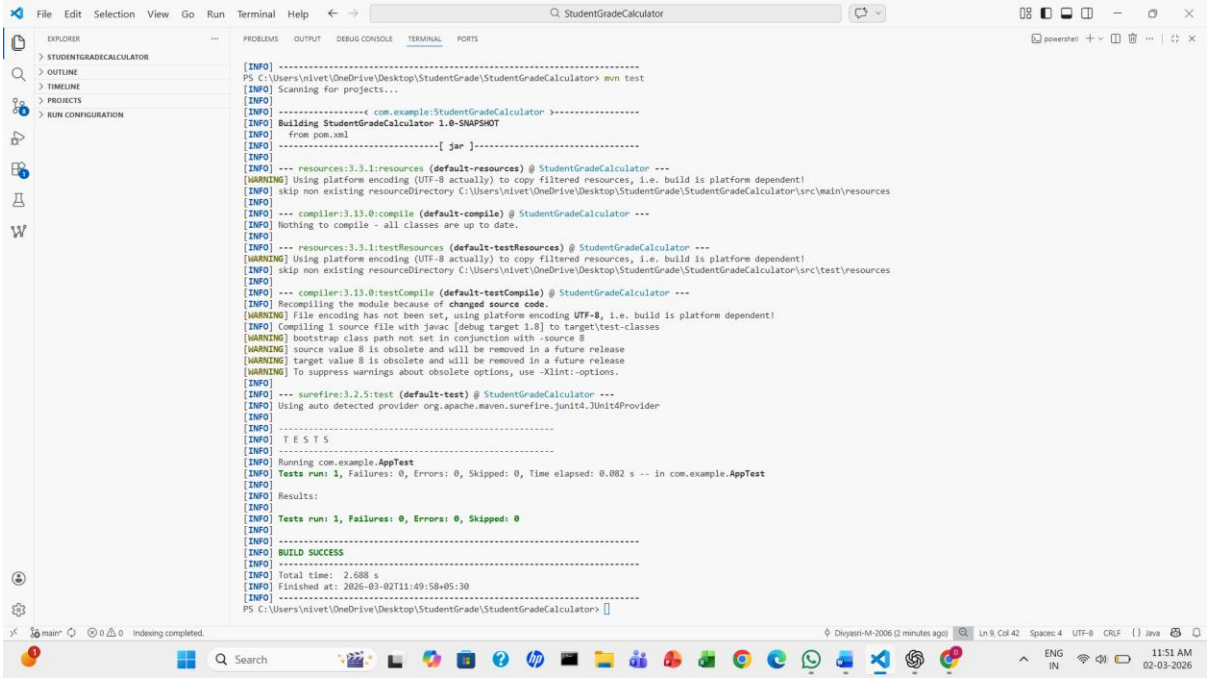
4. Build the Maven Project

- Use Maven commands in Visual Studio Code terminal to:

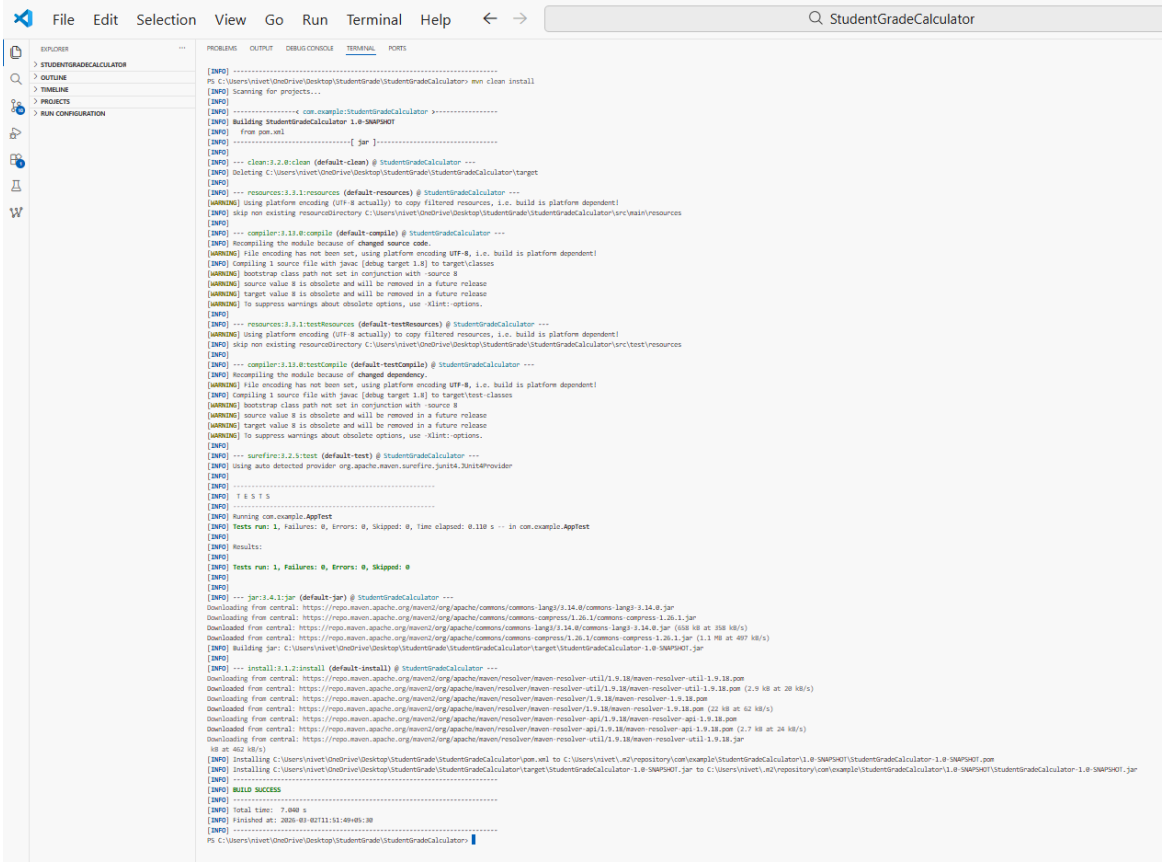
Compile the project



Run tests

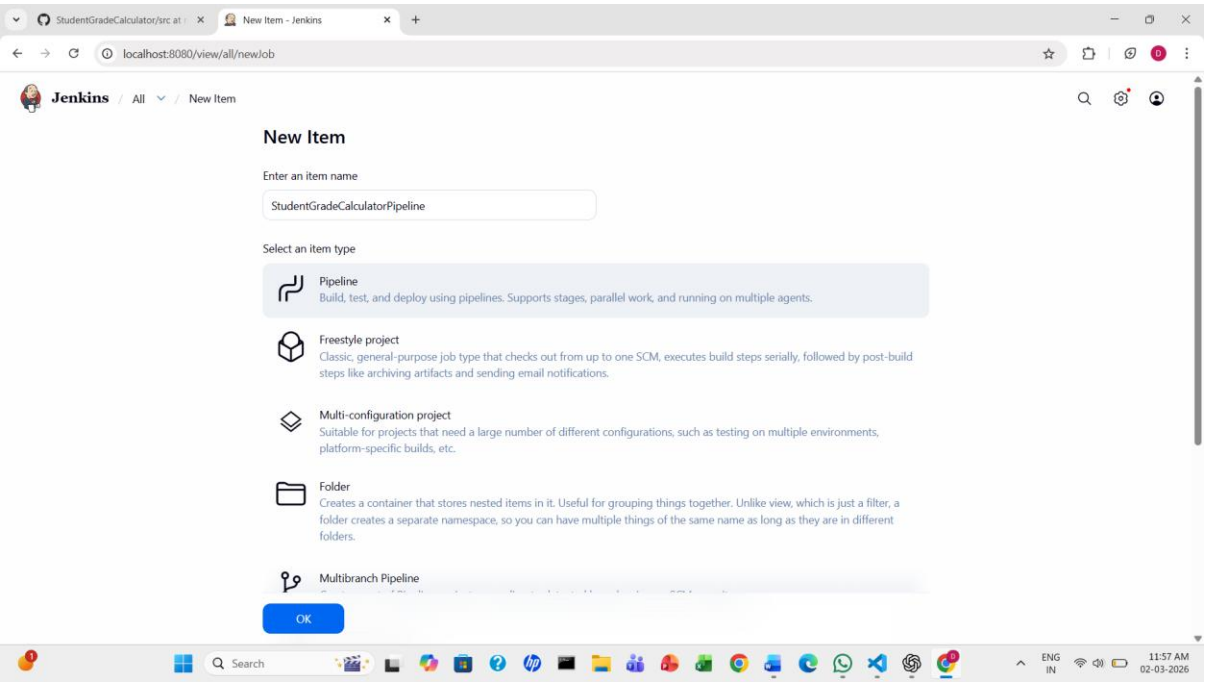


Build the project (mvn clean install)

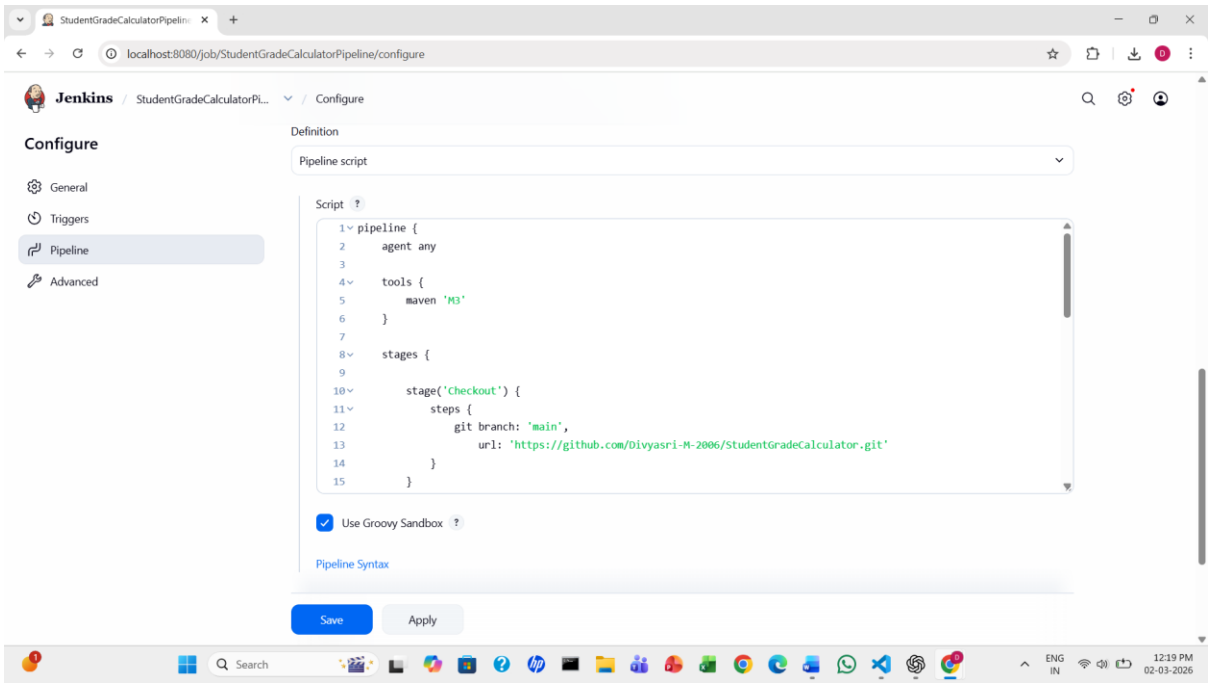


5. Create and Run Jenkins Pipeline

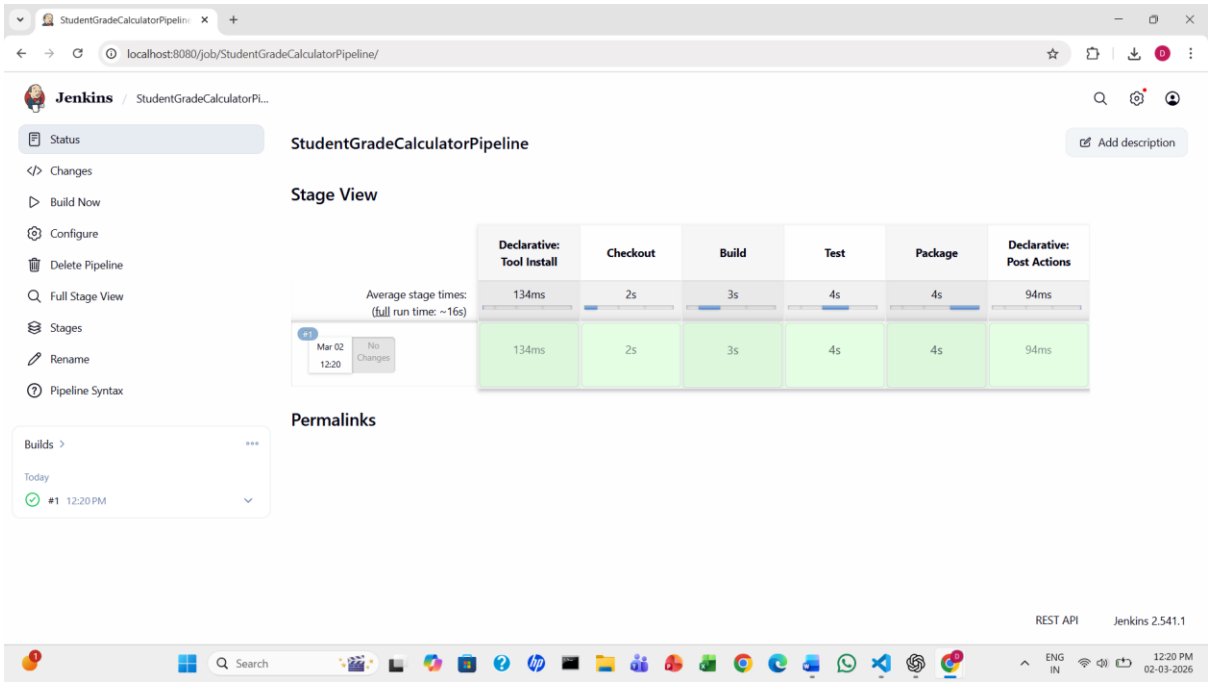
- Create a new Pipeline project in Jenkins.



- Create a pipeline script



- Execute the pipeline and verify successful build completion.



Console - #1 - Jenkins

localhost:8080/job/StudentGradeCalculatorPipeline/1/console

Jenkins / StudentGradeCalculatorPi... / #1

- Status
- Changes
- Console Output
- Edit Build Information
- Delete build '#1'
- Timings
- Git Build Data
- Pipeline Overview
- Restart from Stage
- Replay
- Pipeline Steps
- Workspaces

Console

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```
Started by user Divyasri_M
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\StudentGradeCalculatorPipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Tool Install)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Checkout)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
 > C:\Program Files\Git\cmd\git.exe rev-parse --resolve-git-dir C:\ProgramData\Jenkins\.jenkins\workspace\StudentGradeCalculatorPipeline\.git
# timeout=10
Fetching changes from the remote Git repository
```

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Console - #1 - Jenkins

localhost:8080/job/StudentGradeCalculatorPipeline/1/console

Jenkins / StudentGradeCalculatorPi... / #1

```
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.060 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ StudentGradeCalculator ---
[INFO] Building jar: C:\ProgramData\Jenkins\.jenkins\workspace\StudentGradeCalculatorPipeline\target\StudentGradeCalculator-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.564 s
[INFO] Finished at: 2026-03-02T12:20:28+05:30
[INFO] ------------------------------------------------------------------------
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Pipeline completed successfully!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

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